Practice Quiz: Planning the Creative Process

Part A: Multiple Choice

1. The exploration-exploitation trade-off in creative planning is driven by: A) Budget constraints alone B) The balance between epistemic expected free energy (information gain) and pragmatic expected free energy (goal achievement) C) The number of team members available D) The inventor's personal preference for novelty versus completion

2. The "double diamond" design model represents: A) Two sequential cycles of divergent exploration followed by convergent selection B) Two competing approaches developed simultaneously C) A budget allocation between research and development D) The relationship between cost and quality

3. Creative uncertainty differs from conventional project risk in that: A) It is always smaller than conventional risk B) It involves structural uncertainty — not knowing what you do not know — not just parametric uncertainty C) It can be fully eliminated through better planning D) It only applies to artistic projects, not engineering

4. Stage-gate processes help creative projects by: A) Eliminating all uncertainty before proceeding B) Forcing explicit evidence-based decisions about whether to continue, pivot, or stop C) Replacing creative exploration with linear execution D) Ensuring that no ideas are ever abandoned

5. Information-maximizing milestones prioritize: A) The easiest tasks first to build momentum B) Testing the highest-impact, most-testable uncertainties earliest in the project C) Completing the most expensive components first D) Following the conventional sequence of engineering phases

6. Multi-horizon planning matches planning detail to prediction reliability. This means: A) All horizons should have the same level of detail B) Near-future plans should be specific actions; far-future plans should be directions and visions C) Far-future plans should be more detailed because they are more important D) Only near-future plans are worth creating

7. SpaceX's approach to rocket reuse planning exemplified: A) Traditional waterfall project management B) Fail-fast planning with information-maximizing milestones and adaptive replanning after each attempt C) Ignoring failures and repeating the same approach D) Waiting until all technology was proven before attempting reuse

Part B: Short Answer / Analysis

1. A startup is developing a new type of biodegradable packaging material. They have identified three major uncertainties: (a) whether the material is strong enough for shipping, (b) whether consumers will pay a premium for biodegradable packaging, and (c) whether the manufacturing process can scale to industrial volumes. Using the information-maximizing milestone approach, rank these uncertainties by (impact if wrong) x (testability) and design a three-stage plan that addresses them in the optimal order. Explain why your ordering is superior to addressing them in the original (a, b, c) sequence.

2. An inventor has been exploring alternative designs for a new medical device for six months. She has generated 15 concept sketches, built 4 rough prototypes, and conducted interviews with 12 potential users. Her most recent brainstorming sessions produce concepts very similar to her three leading designs, and two of those three meet all identified constraints. Using the convergence criteria from this module (solution clustering, diminishing returns, constraint satisfaction), advise her on whether to continue exploring or commit to one design. Justify your advice using Active Inference concepts.

3. You are planning an inventive project with a 12-month timeline. Create a multi-horizon plan with three levels of detail: (a) specific daily/weekly actions for the next 2 weeks, (b) monthly objectives for months 2-4, and (c) quarterly direction for months 5-12. For each level, explain why the chosen level of detail is appropriate given the prediction reliability at that horizon. Include at least one planned review point where the plan will be revised based on accumulated evidence.